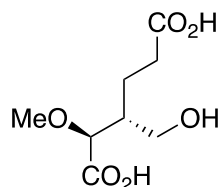


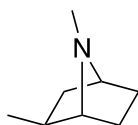
Multi-step Synthesis

1. Design syntheses of the following compounds from starting materials available in the [Sigma catalogue](#) containing six carbons or less. Note that many approaches are possible; strive for an approach with the minimal number of distinct reactions. Draw on your full knowledge of organic reactions. Take your time as these targets are damn *hard*!

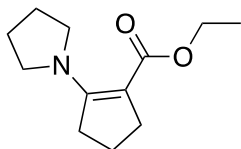
A.



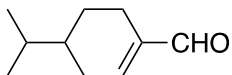
B.



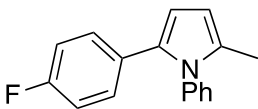
C.



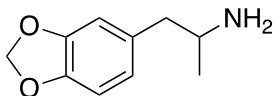
D.



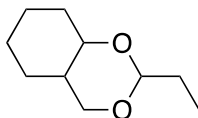
E.



F.



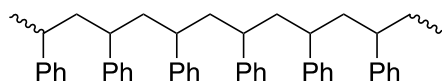
G.



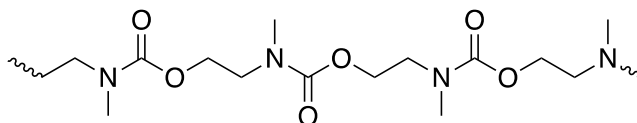
Polymers and Polymerization Reactions

2. For each of the polymers below, draw the structure of a monomer or monomers that can be polymerized to synthesize the polymer.

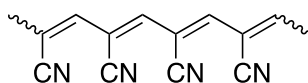
A.



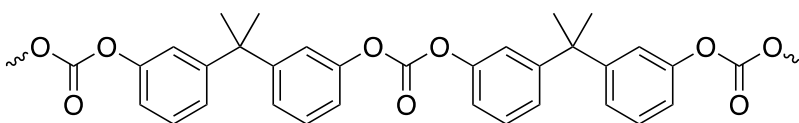
B.



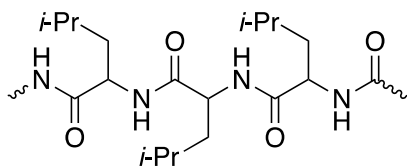
C.



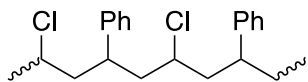
D.



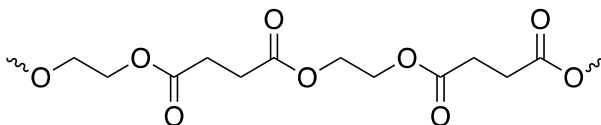
E.



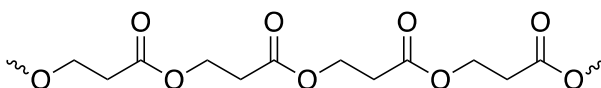
F.



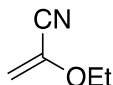
G.



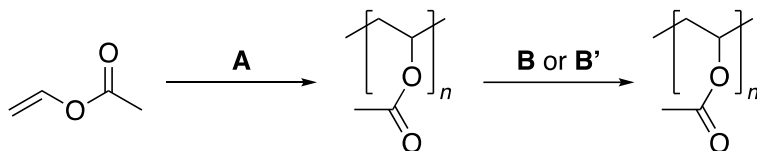
H.



3. The monomer below can be polymerized under either acidic or basic conditions.



- A. Provide a mechanism with curved arrows for polymerization of this monomer in the presence of acid (H^+). Show initiation and addition of two monomer units.
 - B. Provide a mechanism with curved arrows for polymerization of this monomer in the presence of base (OH^-). Show initiation and addition of two monomer units.
4. When 3-methyl-1-butene is treated with a catalytic amount of BF_3 and H_2O , a cationic polymerization process occurs, but the expected homopolymer is not formed. Instead, a random copolymer is obtained containing two distinct repeating units. Provide the structures of the repeating units.
5. Poly(vinyl alcohol) is not made via polymerization of vinyl alcohol. Instead, poly(vinyl acetate) is converted to poly(vinyl alcohol).



- A. Explain why poly(vinyl alcohol) cannot be made from vinyl alcohol.
 - B. Provide a set of reagents **A** that would effectively polymerize vinyl acetate.
 - C. Provide *two* distinct sets of reagents **B** and **B'** for converting poly(vinyl acetate) to poly(vinyl alcohol) that result in different byproducts. *Hint*: sodium acetate is the byproduct of one possible reaction and sodium ethoxide the byproduct of the other.
6. Provide the structure of the polymer expected in each case below and indicate whether each reaction involves chain growth or step growth.

